Euclid, Book I, Proposition I.11

To Draw a Straight Line at Right Angles to a Given Straight Line from a Given Point on It

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1 Euclid's Proposition I.11

Proposition I.11 states:

> To draw a straight line at right angles to a given straight line from a given point on it.

Let AB be the given straight line and let C be the given point on AB.

Euclid's construction proceeds as follows. Choose a point D on AC. By Proposition I.3, construct a point E on the opposite side of C such that

$$CD \cong CE.$$

On the segment DE construct the equilateral triangle DFE by Proposition I.1, and join C to F.

Since

$$CD \cong CE, \qquad DF \cong EF,$$

and CF is common to the two triangles DCF and ECF, Proposition I.8 gives

$$\angle DCF \cong \angle ECF.$$

The points D, C, and E lie on one straight line. Hence the two congruent angles DCF and FCE are adjacent angles formed when the straight line CF stands on the straight line DE.

By Definition 10, each of these angles is therefore a right angle. Consequently, CF is drawn at right angles to the given straight line AB at the given point C.

Thus the required perpendicular has been constructed.

2 Statement

The synthetic content of Proposition I.11 is the existence of a perpendicular through a prescribed point of a given straight line.

Theorem 1 (Euclid I.11)**.** *Let A, C, B be three points such that*

$$A * C * B.$$

Then there exists a point X such that the angle ACX is a right angle.

In the Hilbert development, a right angle is represented directly by the Euclidean definition: an angle is right when it is congruent to its adjacent angle obtained by extending one of its sides through the vertex.

More precisely, we define

$$\text{RightAngle}(A, O, B)$$

to mean that there exists a point C such that

$$A * O * C$$

and

$$\angle AOB \cong \angle BOC.$$

Thus Proposition I.11 is represented formally by

$$A * C * B \quad \Longrightarrow \quad \exists X \;\; \text{RightAngle}(A, C, X).$$

The corresponding Lean definition is

```
def HilbertRightAngle
(Geo : Geometry.Geo)
(A O B : Geo.Point) : Prop :=
exists C : Geo.Point,
Geo.Between A O C /\
Geo.AngleCongruent A O B B O C
```

and the proposition itself has the form

```
theorem euclid_proposition_11
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X
```

Notice that the formal statement does not attempt to encode Euclid's particular construction of the perpendicular. It expresses the geometric result of that construction: through the prescribed point C on the given line there exists a ray CX forming a right angle with the line.

3 Proof

Let AB be the given straight line and let C be the prescribed point on it.

Choose a point D on AC. By Proposition I.3, cut off from the ray starting at C in the direction of B a segment CE congruent to CD. Thus

$$CD \cong CE,$$

and the points D, C, E lie on one straight line, with C between D and E.

Construct the equilateral triangle DFE on the segment DE by Proposition I.1, and join C to F. Since the triangle is equilateral,

$$DF \cong EF.$$

Moreover,

$$CF \cong CF$$

by reflexivity.

Consider the triangles

$$\triangle DCF \qquad \text{and} \qquad \triangle ECF.$$

Their three corresponding sides satisfy

$$CD \cong CE, \qquad DF \cong EF, \qquad CF \cong CF.$$

Therefore Proposition I.8 yields

$$\angle DCF \cong \angle ECF.$$

Since D, C, E are collinear and C lies between D and E, the rays CD and CE are opposite rays. Hence

$$\angle DCF \qquad \text{and} \qquad \angle FCE$$

are adjacent angles formed by the straight line CF standing on the straight line DE.

The two adjacent angles are congruent. By Euclid's Definition 10, each of them is therefore a right angle.

Consequently,

$$\angle DCF$$

is a right angle, and the straight line through C and F is perpendicular to the given straight line AB at the prescribed point C.

This proves Proposition I.11.

4 Implementation Architecture

The formal reconstruction separates the notion of a right angle from the particular statement of Proposition I.11.

At the foundational level, the predicate `HilbertRightAngle` expresses Euclid's Definition 10. An angle AOB is right when there exists a point C on the extension of AO through O such that

$$\angle AOB \cong \angle BOC.$$

The substantial geometric result is then established independently as

```
hilbert_right_angle_exists
```

which proves that, whenever

$$A * C * B,$$

there exists a point X such that

$$\text{HilbertRightAngle}(A, C, X).$$

The proof of this existence theorem is considerably more elaborate than the final formalization of Proposition I.11. It uses the developed Hilbert theory of angle construction, segment construction, midpoints, triangle congruence, ray order, plane separation, and angle addition.

Once this general result has been established, Proposition I.11 itself becomes a direct interface theorem:

```
hilbert_right_angle_exists
|
v
euclid_proposition_11
```

More explicitly, the architecture is

$$\boxed{\texttt{HilbertRightAngle}}$$
$$\downarrow$$
$$\boxed{\text{Hilbert synthetic geometry}}$$

angle construction
segment construction
midpoint existence
Hilbert SSS
SameSide/OppositeSide theory
ray-order theorem
angle addition

$$\downarrow$$
$$\boxed{\texttt{hilbert\_right\_angle\_exists}}$$
$$\downarrow$$
$$\boxed{\texttt{euclid\_proposition\_11}}$$

Thus the formal dependency structure differs sharply from Euclid's historical construction. Euclid obtains the perpendicular by constructing an equilateral triangle and applying Proposition I.8. The Hilbert development instead proves a general right-angle existence theorem from the already developed synthetic theory and then obtains Proposition I.11 as an immediate consequence.

This distinction is intentional. The proposition file records the mathematical content of Euclid I.11, while the underlying Hilbert layer is free to establish that content by a more general synthetic argument.

5 Hilbert Reconstruction

The Hilbert proof of the existence theorem does not reproduce Euclid's equilateral-triangle construction. Instead, it derives a right angle from the general machinery already available in the Hilbert layer.

Assume

$$A * C * B.$$

Thus A, C, B lie on a common line, which we denote by b.

Choose a point D outside b. The angle-construction theorem is then applied at C to construct a point E, on the same side of b as D, such that

$$\angle ACD \cong \angle BCE.$$

The uniqueness part of angle construction determines the ray CE.

Next choose a point F on the ray CD such that

$$CF \cong CE.$$

Let X be the midpoint of FE. Hence

$$F * X * E$$

and

$$FX \cong XE.$$

Consider the triangles

$$\triangle CFX \qquad \text{and} \qquad \triangle CEX.$$

They satisfy

$$CF \cong CE, \qquad FX \cong XE, \qquad CX \cong CX.$$

Hilbert SSS therefore gives

$$\angle FCX \cong \angle ECX.$$

Since F and D lie on the same ray from C, replacing the ray CF by the ray CD does not change the corresponding angle. Consequently,

$$\angle DCX \cong \angle ECX.$$

We have therefore obtained the two component congruences

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX.$$

The remaining problem is to justify that these component angles may be added in the required configuration.

This is the technically most substantial part of the reconstruction. The ray-order theorem applied to the noncollinear configuration determines two possible cases:

$$\text{RayMeetsSegment}(CD, EA)$$

or

$$\text{RayMeetsSegment}(CE, DA).$$

In the first case, the ray CD meets the open segment EA at a point Y. The SameSide/OppositeSide calculus shows that

$$A \text{ and } X$$

lie on opposite sides of the line CD, while

$$B \text{ and } X$$

lie on opposite sides of the line CE.

In the second case, the ray CE meets the open segment DA at a point Y. Plane-separation arguments show instead that

$$A \text{ and } X$$

lie on the same side of CD, while

$$B \text{ and } X$$

lie on the same side of CE.

Thus in both cases the two configurations have the same side pattern:

$$\text{SameSide}(A, X; CD) \iff \text{SameSide}(B, X; CE).$$

The proof also establishes that X does not lie on the original line AB. Hence both

$$\angle ACX \qquad \text{and} \qquad \angle BCX$$

are nondegenerate angles.

The angle-addition theorem can now be applied to

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX.$$

It yields

$$\angle ACX \cong \angle BCX.$$

Since $A * C * B$, the rays CA and CB are opposite. Reversing the order of the sides in the second angle gives

$$\angle ACX \cong \angle XCB.$$

Therefore the two adjacent angles formed by CX with the straight line AB are congruent.

By the definition of `HilbertRightAngle`,

$$\text{HilbertRightAngle}(A, C, X).$$

Hence

$$\exists X \ \text{HilbertRightAngle}(A, C, X),$$

which is the required Hilbert form of Proposition I.11.

6 Lean Formalization

The formalization is divided into three levels: the definition of a right angle, the general Hilbert existence theorem, and the final Euclid I.11 interface theorem.

The notion of a right angle is represented by the following definition:

```
def HilbertRightAngle
(Geo : Geometry.Geo)
(A O B : Geo.Point) : Prop :=
exists C : Geo.Point,
Geo.Between A O C /\
Geo.AngleCongruent A O B B O C
```

This is a direct formalization of the adjacent-angle characterization of a right angle. The witness C extends the ray OA through O, and the congruence

$$\angle AOB \cong \angle BOC$$

states that the two adjacent angles are equal.

The main work is contained in the general existence theorem:

```
theorem hilbert_right_angle_exists
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X := by
...
```

The proof first extracts the incidence and nondegeneracy information contained in `hACB`. It then chooses an auxiliary point off the line AB and constructs a congruent angle on the opposite ray at C.

A point F is subsequently constructed on the auxiliary ray with

$$CF \cong CE,$$

and the midpoint X of FE is introduced. The central triangle congruence step is encoded by an application of Hilbert SSS:

```
have hSSS :=
HilbertSSS ...
```

This produces the congruence of the two angles at C required for the second component of the final angle-addition argument.

The principal technical issue is not the congruence calculation but the relative position of the rays. The proof therefore invokes the ray-order theorem and splits into its two possible cases:

```
rcases hRayOrder with hCaseD | hCaseE
```

Each branch establishes the same logical condition required by the angle-addition theorem:

```
have hSideConfiguration :
HilbertSameSide Geo A X lineCD <->
HilbertSameSide Geo B X lineCE := by
...
```

In one branch both propositions are false; in the other both are true. This case distinction isolates the plane-separation reasoning from the final angle calculation.

After proving the necessary noncollinearity conditions, the two component angle congruences are combined by

```
have hACX_BCX :
Geo.AngleCongruent A C X B C X :=
hilbert_angle_addition
...
hSideConfiguration
...
hAngleE
hDCX_ECX
```

The second angle is then reversed:

```
have hACX_XCB :
Geo.AngleCongruent A C X X C B :=
(Geo.angle_congruent_reverse_second
A C X B C X).mp hACX_BCX
```

Since the original hypothesis already supplies

$$A * C * B,$$

the point B itself is the witness required by the definition of `HilbertRightAngle`. The existence theorem is therefore completed by

```
refine <X, ?_>
refine <B, hACB, ?_>
exact hACX_XCB
```

Once this theorem is available, the formal statement of Proposition I.11 contains no additional geometric argument:

```
theorem euclid_proposition_11
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X := by
exact
hilbert_right_angle_exists
Geo A C B hACB
```

Thus the Lean theorem corresponding to Euclid I.11 is deliberately small. The substantial synthetic argument has been factored into `hilbert_right_angle_exists`, making right-angle existence available independently of Proposition I.11 and reusable in the subsequent development.

7 Discussion

Proposition I.11 provides an instructive example of the distinction between the organization of Euclid's *Elements* and the organization of the Hilbert formalization.

In Euclid's development, Proposition I.11 is itself a construction. Its proof depends explicitly on earlier propositions:

$$\text{I.1}, \qquad \text{I.3}, \qquad \text{I.8},$$

together with the definition of a right angle. The perpendicular is obtained by constructing an equilateral triangle and comparing two triangles by SSS.

The formal Hilbert development reaches the same result through a different dependency structure. Proposition I.11 does not carry the construction itself. Instead, the substantial argument is isolated as the general theorem

```
hilbert_right_angle_exists
```

and Proposition I.11 merely exposes this theorem at the Euclidean interface.

This is similar in form to Proposition I.10. In both cases the final Euclid theorem is short because the required existence result has already been established in the underlying Hilbert theory. The amount of work hidden behind the interface, however, is very different.

For Proposition I.10, midpoint existence was already a natural primitive result of the developed theory. Proposition I.11 required the introduction of a new general notion,

```
HilbertRightAngle
```

and a substantial synthetic existence proof.

The proof also reveals where the formal difficulty of the result lies. The elementary congruence argument is comparatively short: after constructing F and the midpoint X of FE, SSS gives

$$\angle FCX \cong \angle ECX.$$

The difficult part is proving that the component angles occur in the correct relative configuration so that angle addition is legitimate.

This requires the developed theory of

$$\text{SameSide}, \qquad \text{OppositeSide}, \qquad \text{rays}, \qquad \text{betweenness},$$

together with the ray-order theorem. The resulting two-case analysis is substantially longer than the final congruence calculation.

Thus Proposition I.11 illustrates an important feature of the formalization. Statements that are elementary in an informal diagrammatic proof may require a significant amount of explicit order and separation theory once all geometric configurations must be justified formally.

At the same time, this complexity is not local overhead that must be repeated in later propositions. The result has been packaged as `hilbert_right_angle_exists`. Subsequent arguments may use right-angle existence directly without reconstructing the SameSide/OppositeSide analysis.

The resulting architecture is therefore

$$\text{foundational Hilbert theory} \longrightarrow \text{general synthetic lemmas} \longrightarrow \text{Euclid propositions.}$$

Proposition I.11 is a particularly clear instance of this pattern: its Euclidean interface is almost trivial precisely because the geometric content has been absorbed into the reusable Hilbert layer.

Finally, the comparison with Euclid should not be interpreted as a requirement that the formal proof reproduce Euclid's historical proof step by step. The purpose of the proposition layer is to recover the mathematical statements of Book I over the developed Hilbert theory. When that theory already supplies a stronger or more reusable result, the Euclid proposition can appropriately appear as a thin interface theorem.